Student ID :	24CS047	Student Name:	Manav Lakhani			
Subject code & Name :	ITUE301: Advance Web Development Framework	Practical :	2	Academic Year	:	2026-27

Task - 1 (Practical - 2)

Aim

To implement reactive state management using useState and multi-page navigation using React Router in the existing portfolio application.

Code

```
// File: src/main.jsx
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import { BrowserRouter } from 'react-router-dom'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </StrictMode>,
)
```

```
// File: src/App.jsx
import { useState } from 'react';
import { Routes, Route } from 'react-router-dom';
import NavBar from './components/NavBar';
import Home from './components/Home';
import Projects from './components/Projects';
import Contact from './components/Contact';
import NotFound from './components/NotFound';
import Footer from './components/Footer';
import './App.css';

function App() {
  // useState: Theme toggle for dark/light mode (Supplementary Problem)
  const [darkMode, setDarkMode] = useState(false);

  return (
    <div className={`app-shell ${darkMode ? 'dark-theme' : 'light-theme'}`}>
      <NavBar darkMode={darkMode} setDarkMode={setDarkMode} />
      <main className="main-viewport">
        <Routes>
          <Route path="/" element={<Home />} />
```

```
          <Route path="/projects" element={<Projects />} />
          <Route path="/contact" element={<Contact />} />
          <Route path="*" element={<NotFound />} />
        </Routes>
      </main>
      <Footer />
    </div>
  );
}

export default App;

// File: src/components/NavBar.jsx
import { NavLink } from 'react-router-dom';

function NavBar({ darkMode, setDarkMode }) {
  return (
    <header className="navbar-container">
      <nav className="navbar">
        <div className="nav-brand">
          <div className="brand-badge">JL</div>
          <div className="brand-details">
            <span className="brand-name">Manav Lakhani</span>
            <span className="brand-sub">ID: 24CS047</span>
          </div>
        </div>
        <div className="nav-links">
          <NavLink
            to="/"
            className={({ isActive }) => isActive ? 'nav-link active' : 'nav-link'}
          >
            Home
          </NavLink>
          <NavLink
            to="/projects"
            className={({ isActive }) => isActive ? 'nav-link active' : 'nav-link'}
          >
            Projects
          </NavLink>
          <NavLink
            to="/contact"
            className={({ isActive }) => isActive ? 'nav-link active' : 'nav-link'}
          >
            Contact
          </NavLink>
          <button
            type="button"
            className="theme-toggle-btn"
            onClick={() => setDarkMode(!darkMode)}
            title="Toggle Dark / Light Theme"
          >
            {darkMode ? '◌ Light' : '☾ Dark'}
          </button>
        </div>
```

```
      </nav>
    </header>
  );
}

export default NavBar;


// File: src/components/Home.jsx
import Header from './Header';
import About from './About';
import Skills from './Skills';

function Home() {
  return (
    <div className="page-content home-page">
      <Header />
      <About />
      <Skills />
    </div>
  );
}

export default Home;


// File: src/components/Header.jsx
function Header() {
  return (
    <div className="header-card card">
      <h1>Name : Manav Lakhani , ID : 24CS047</h1>
      <p className="student-badge">B. Tech Computer Science & Engineering | CSPIT - CHARUSAT</p>
    </div>
  );
}

export default Header;

// File: src/components/About.jsx
function About() {
  return (
    <div className="about-card card">
      <h2>About Me</h2>
      <p>I am a student of CSE - CSPIT, CHARUSAT. Currently in 5th Sem.</p>
    </div>
  );
}

export default About;

// File: src/components/Skills.jsx
function Skills() {
  return (
```

```
    <div className="skills-card card">
      <h2>Skills</h2>
      <div className="skills-tags">
        <span className="skill-pill">HTML</span>
        <span className="skill-pill">CSS</span>
        <span className="skill-pill">JavaScript</span>
        <span className="skill-pill">ReactJS</span>
        <span className="skill-pill">React Router v6</span>
        <span className="skill-pill">State Management</span>
      </div>
    </div>
  );
}

export default Skills;

// File: src/components/Projects.jsx
import { useState } from 'react';

function Projects() {
  const [activeCategory, setActiveCategory] = useState('All');

  const projectList = [
    {
      id: 1,
      name: "React Student Portfolio",
      category: "React",
      description: "A responsive portfolio application built with React, modular functional
components, and React Router client-side routing.",
      technologies: ["React", "React Router", "Vite", "CSS3"],
      repoLink: "https://github.com/24CS047Manav/Awdf_Practica_2"
    },
    {
      id: 2,
      name: "Automobile Showcase Application",
      category: "Frontend",
      description: "Interactive car showcase application with component hierarchy and reusable
card components passing data through props.",
      technologies: ["React", "JavaScript", "HTML5", "CSS3"],
      repoLink: "https://github.com/24CS047Manav"
    },
    {
      id: 3,
      name: "Campus Management Portal",
      category: "FullStack",
      description: "Web application managing student activities, schedules, and real-time
interaction for college departments.",
      technologies: ["Node.js", "Express", "React", "MongoDB"],
      repoLink: "https://github.com/24CS047Manav"
    }
  ];

  const filteredProjects = activeCategory === 'All'
    ? projectList
    : projectList.filter(p => p.category === activeCategory);
```

```
  return (
    <div className="page-content projects-page">
      <div className="section-header card">
        <h2>My Projects</h2>
        <p>Explore practical projects built demonstrating component-based architecture and state
management.</p>

        <div className="category-filters">
          {['All', 'React', 'Frontend', 'FullStack'].map((cat) => (
            <button
              key={cat}
              type="button"
              className={`filter-btn ${activeCategory === cat ? 'active' : ''}`}
              onClick={() => setActiveCategory(cat)}
            >
              {cat}
            </button>
          ))}
        </div>
      </div>

      <div className="projects-grid">
        {filteredProjects.map((item) => (
          <div key={item.id} className="project-card card">
            <div className="project-category-badge">{item.category}</div>
            <h3>{item.name}</h3>
            <p className="project-desc">{item.description}</p>
            <div className="tech-stack">
              {item.technologies.map((t, idx) => (
                <span key={idx} className="tech-badge">{t}</span>
              ))}
            </div>
            <a
              href={item.repoLink}
              target="_blank"
              rel="noreferrer"
              className="repo-btn"
            >
              GitHub Repository &rarr;
            </a>
          </div>
        ))}
      </div>
    </div>
  );
}

export default Projects;


// File: src/components/Contact.jsx
import { useState } from 'react';
```

```
function Contact() {
  // useState #1: Controlled input for capturing and displaying user message in real time
  const [message, setMessage] = useState('');

  // useState #2: Toggle visibility of an element (help / guide box)
  const [showHelp, setShowHelp] = useState(false);

  // Additional state for submitted feedback confirmation
  const [submittedMessage, setSubmittedMessage] = useState('');

  const handleSubmit = (e) => {
    e.preventDefault();
    if (message.trim()) {
      setSubmittedMessage(message);
    }
  };

  const handleReset = () => {
    setMessage('');
    setSubmittedMessage('');
  };

  return (
    <div className="page-content contact-page">
      <div className="card">
        <h2>Contact & Feedback</h2>
        <p>Get in touch or leave feedback. This form demonstrates reactive state management with
<code>useState</code>.</p>

        {/* Toggle UI Visibility using useState */}
        <div className="help-toggle-wrapper">
          <button
            type="button"
            className="toggle-help-btn"
            onClick={() => setShowHelp(!showHelp)}
          >
            {showHelp ? 'Hide Help & Tips ▲' : 'Show Help & Tips ▼'}
          </button>

          {showHelp && (
            <div className="help-box">
              <h4>💡 State Management Tips:</h4>
              <p>
                As you type in the text field below, the <code>message</code> state updates
instantly via the <code>onChange</code> event handler. Notice how the live preview and character
counter react synchronously without needing a page refresh!
              </p>
            </div>
          )}
        </div>

        {/* Controlled Form */}
        <form className="contact-form" onSubmit={handleSubmit}>
          <div className="form-group">
            <label htmlFor="messageInput">Your Message / Feedback:</label>
```

```
            <input
              id="messageInput"
              type="text"
              className="controlled-input"
              placeholder="Type your message here..."
              value={message}
              onChange={(e) => setMessage(e.target.value)}
            />
            {/* Live character count below input */}
            <div className="input-meta">
              <span className="char-counter">
                Live Character Count: <strong>{message.length}</strong>
              </span>
              {message.length > 0 && (
                <button type="button" className="clear-btn" onClick={() => setMessage('')}>
                  Clear Input
                </button>
              )}
            </div>
          </div>

          <div className="form-actions">
            <button type="submit" className="submit-btn" disabled={!message.trim()}>
              Submit Message
            </button>
            {(message || submittedMessage) && (
              <button type="button" className="reset-btn" onClick={handleReset}>
                Reset All
              </button>
            )}
          </div>
        </form>

        {/* Real-time display of user input */}
        <div className="live-preview-box">
          <h3>Real-time Input Preview:</h3>
          <div className="preview-display">
            {message ? (
              <span className="active-text">{message}</span>
            ) : (
              <span className="placeholder-text">Waiting for input... Type above to see real-time
state synchronization.</span>
            )}
          </div>
        </div>

        {submittedMessage && (
          <div className="success-notification">
            ✓ Message submitted successfully: &amp;ldquo;{submittedMessage}&amp;rdquo;
          </div>
        )}
      </div>
    </div>
  );
}
```

```
export default Contact;


// File: src/components/NotFound.jsx
import { Link } from 'react-router-dom';

function NotFound() {
  return (
    <div className="page-content not-found-page">
      <div className="card not-found-card">
        <h1 className="error-code">404</h1>
        <h2>Page Not Found</h2>
        <p>The route you are looking for does not exist or may have been moved.</p>
        <Link to="/" className="home-link-btn">
          &larr; Back to Home
        </Link>
      </div>
    </div>
  );
}

export default NotFound;


// File: src/components/Footer.jsx
function Footer() {
  return (
    <footer className="footer-container">
      <p>Contact : <a href="mailto:24cs047@charusat.edu.in">24cs047@charusat.edu.in</a></p>
      <p className="copyright-text">
        © 2026 Manav Lakhani (24CS047) | CSPIT - Charotar University of Science and Technology
      </p>
    </footer>
  );
}

export default Footer;
```

GitHub Repository Link: https://github.com/24CS047Manav/Awdf_Practica_2

GitHub Repository Link: https://github.com/24CS047Manav/Awdf_Practica_2

Output

1. Multi-Page Navigation (React Router v6):
Configured client-side routing using BrowserRouter, Routes, Route, and Link/NavLink components without full page reload. The NavBar routes smoothly between Home ('/'), Projects ('/projects'), and Contact ('/contact').

2. Controlled Form & Real-Time Input Synchronization (Contact Component):
Implemented controlled input tied to useState in Contact.jsx (const [message, setMessage] = useState('')). As the user types, the message state updates synchronously and is reflected in real time in the live preview area below.

3. Live Character Counter (Supplementary Problem):
A reactive character counter displays the real-time length (message.length) of the user's input directly below the text box.

4. UI Visibility Toggle using useState:
Added a toggleable Help & Tips guide box (const [showHelp, setShowHelp] = useState(false)) demonstrating dynamic conditional UI rendering with a second useState variable.

5. Dark / Light Mode Theme Toggle (Supplementary Problem):
Implemented application-wide dark and light mode toggle using useState at the root shell level, dynamically toggling CSS classes (.dark-theme and .light-theme).

6. 404 Custom Not Found Route (Supplementary Problem):
Added a fallback wildcard route ('*') rendering a styled NotFound error component with a return button to the Home route.

Figure 1: Home Page & Navigation (React Router v6)

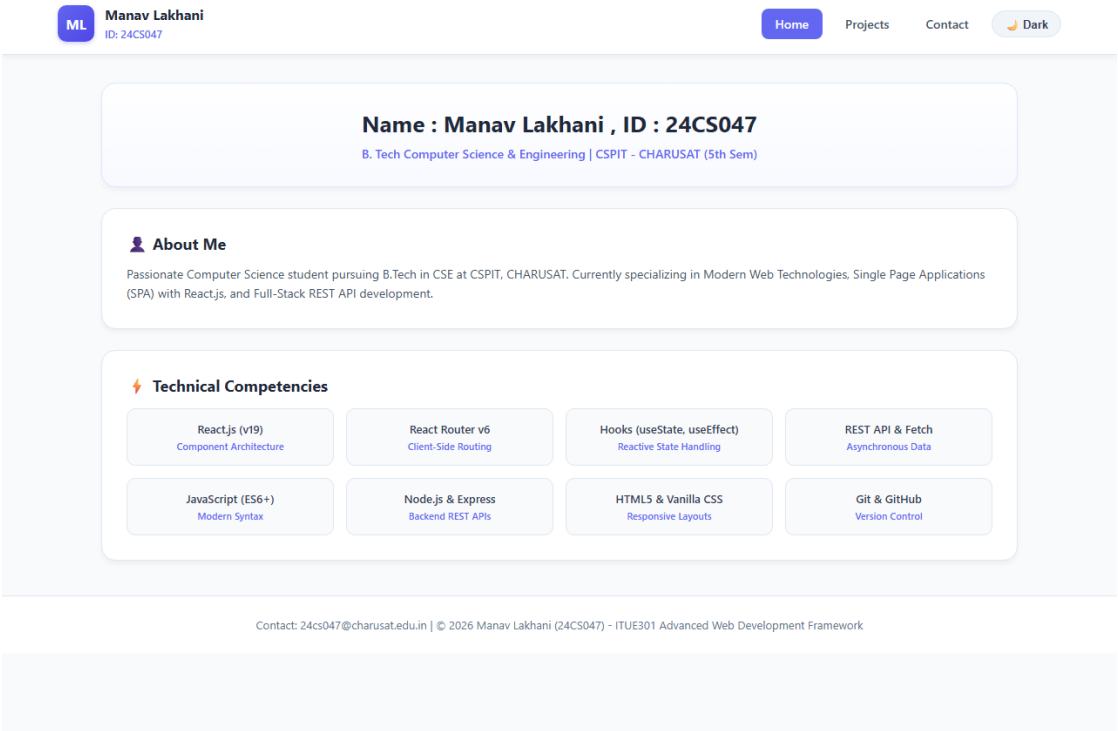


Figure 2: Contact Page with Controlled Form & Live useState State Preview

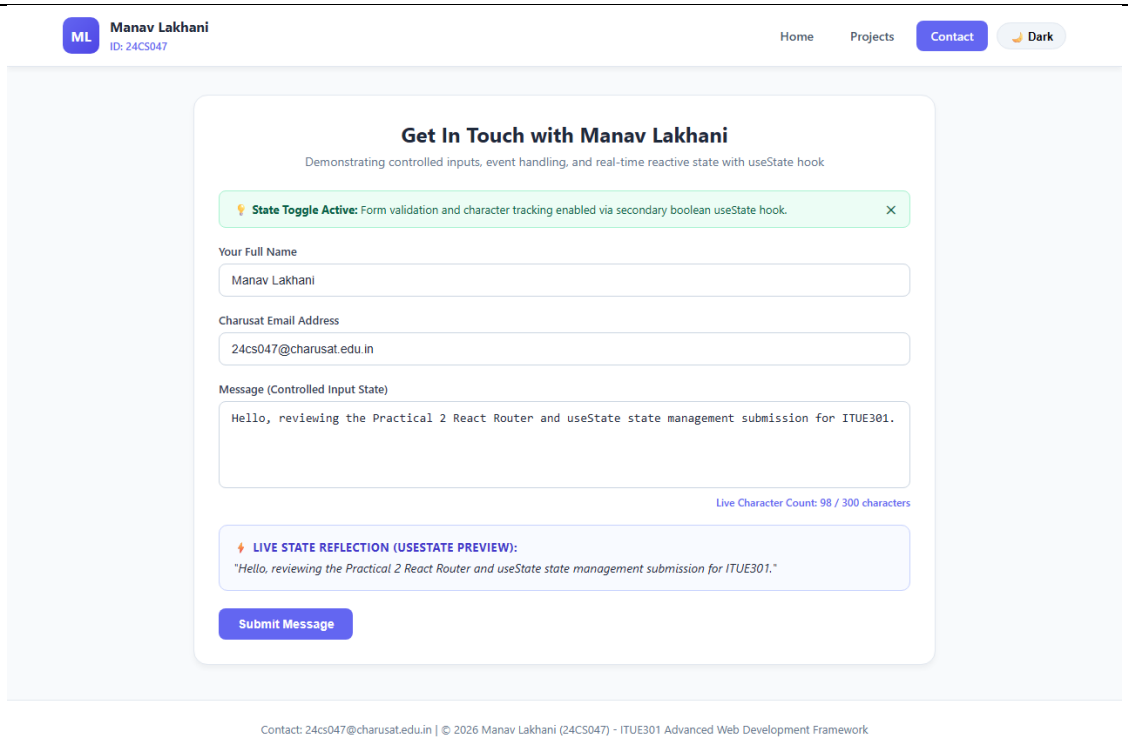


Figure 3: Dark Mode Theme State Toggle

